

ENBC332: Statistics, Data Analysis, and Data Visualization

Instructions for the Project

Fall 2025

Overall points: 100 (20% of total grade)

Due dates:

- **Project proposal:** Wednesday, September 24, 2025, by the end of the day
- **Dropping the group members:** Not Applicable for individual projects
- **Completed project:** Wednesday, December 3, 2025, by the end of the day
- **Presentations:** Wednesday, December 10, 2025, in the classroom BSE4321

Note: Late submission is allowed up to 24 hours after the due date, but a 50% penalty will be applied.

Note: Please aim to write clean report for your project. Also, aim to write clean, well-organized code that is easy for others to understand. Be sure to include comments or explanations where needed. Failure to do so may result in a deduction of points.

Note: You may submit multiple codes. Please provide a clear explanation of the purpose and functionality of each one.

Note: Please name your files using the format: "**name_filename.R**" or "**name_filename.ipynb**". E.g., if your code performs ANOVA test, name your code "**Azarnoosh_ANOVA.R**".

Note: Please submit a **single ZIP file** named "**name.zip**" containing the following:

- The dataset you used
- R code files
- Project in PDF as well as Word document, or LaTeX
- Presentation in pptx and PDF format

General Description of the Project

For the data analysis project, you will explore questions that intrigue you using the statistical methods we cover in ENBC332: Statistics, Data Analysis, and Data Visualization. The project requires you to integrate all the material covered in this course. A good project begins with a well-defined **hypothesis**. Start with questions that interest you, then think about how to collect and analyze data to answer them. Make sure your chosen topic is relevant to biomedical studies. Avoid broad topics, as they lead to weaker projects. Instead, focus on specific hypotheses and design your analysis to test them.

Project Proposal

You should **SUBMIT A PROJECT PROPOSAL** by the proposal due date mentioned above. The proposal should be just **ONE PAGE** and outline your intended project. Be as specific as possible, detailing the question you wish to investigate and your general approach to data collection. The instructor will provide feedback on the proposal topic. The more comprehensive your proposal, the more valuable the feedback you will receive! Your proposal should address the following questions:

- What is the topic of your project?
- What key issues or problems do you plan to tackle?
- What is your hypothesis statement?
- Describe the data you intend to use or collect, including the variables you will analyze. You don't need to provide an in-depth data analysis; detailed plans will be submitted on the final written project.
- What questions or concerns do you have regarding your project?

Please note that the project proposal is not graded; it is primarily intended to help you receive feedback on your project idea. It is important to begin working on the project as soon as possible, especially in terms of sourcing data and gathering background information.

Project Report

You are required to gather data from online sources, ensuring that you carefully select and organize the dataset yourself. For this project, your analysis must be based on data you collect from online resources such as:

1. Title Page:
 - Project title
 - Your name
 - Course name and code
 - Date
2. Abstract:
 - A brief summary of the project, including the research question, methods, results, and conclusions.
3. Introduction (half a page is sufficient):
 - A brief Background information on the topic.

- Explanation of the significance of the research question.
 - Objectives of the study.
- 4. Methodology:
 - Description of the data collection process.
 - Explanation of the statistical methods used for analysis.
- 5. Results:
 - Presentation of the findings, including tables, graphs, and any relevant statistics.
- 6. Discussion:
 - Interpretation of the results.
 - Implications of the findings and how they relate to the research question.
- 7. Conclusion
 - Summary of key findings.
 - Recommendations for future research or practical applications.
- 8. References (if applicable)
 - List of all sources cited in the report, formatted according to a specific citation style (e.g., APA, MLA).
- 9. Appendices (if applicable)
 - Additional materials such as raw data, supplementary analyses, or detailed calculations.

Formatting Guidelines:

- **Font & Size:** Arial 11pt
- **Line Spacing:** 1.5 spacing for readability
- **Margins:** 1-inch on all sides
- **Page Numbers:** Bottom center
- **Headings & Subheadings:** Consistent formatting (bold or large font), numbered if needed
- **Figures & Tables:** Labeled properly with captions and referenced in text
- **Equations:** Numbered and formatted consistently

Citation & Referencing:

- **Citation Style:** APA
- **In-text Citations:** Properly placed throughout the document
- **Bibliography:** Complete and formatted correctly

Submission Format:

- **File Type:** PDF file as well as Word document, or LaTeX
- **Length Limit:** 20 pages
- **Naming Convention:** " name_filename.py" or " name.zip"
- **Submission Method:** Online via ELMS/Canvas

Presentation

In your presentation, you should briefly introduce your research question and dataset, explain the statistical methods you applied and why they were appropriate, and present your key findings with clear visuals such as graphs or tables. Be sure to interpret your results in context, noting any limitations, and conclude with the main takeaways. Your presentation should be concise, well-organized, and within the given time, followed by a short Q&A.

Data Collection

The key to any statistical analysis is clearly defining questions and collecting relevant data. You will gather data from online sources, but you must select and organize the dataset yourself. For this project, you must analyze data **you collect from online resources** such as:

- [Kaggle](#)
- [Google Dataset Search](#)
- [UCI Machine Learning Repository](#)
- [Data.gov](#)
- [AWS Open Data Registry](#)
- [OpenML](#)
- [FiveThirtyEight](#)
- [The World Bank Open Data](#)
- [Figure Eight Datasets](#)
- [Quandl](#)
- [Data World](#)
- [Awesome Public Datasets](#)

Some ideas for projects

Here are some project ideas in the area of bioengineering that would allow you to apply statistical techniques:

- Analysis in cancer patients:
 - Lung cancer
 - Breast cancer
 - Prostate cancer
 - Skin cancer
 - Pancreatic Cancer
 - Ovarian Cancer
 - Leukemia
 - Cervical Cancer
 - Liver Cancer
 - Stomach (Gastric) Cancer
- Analysis of clinical trial data involving a medical device or drug
- Biomarker discovery to identify potential biomarkers for diagnosing a disease
- Biomechanics and motion analysis on human joint movement or muscle activity

- Drug dosage optimization: Analyze the relationship between drug dosage and patient outcomes to optimize treatment
- Survival analysis in biomedical studies
- Design of experiments for medical device testing

Grading Breakdown:

- Project Report: 90% of the total grade.
- Presentation: 10% of the total grade.

Project Grading Guidelines

Project Report (90%): The project report will be evaluated based on your analysis, methodology, and conclusions. The report should be well-structured, clear, and logically organized. Use of proper formatting, grammar, and citations where appropriate.

Your project report will be graded based on the following criteria:

Problem Definition (10%):

- Clear and concise explanation of the research question or problem being analyzed.
- Relevance of the problem to the course material.

Data Collection and Description (10%):

- Description of the dataset used, including source, size, and key variables.
- Justification for choosing the dataset.
- Any data cleaning or preprocessing steps clearly outlined. For example, the dataset contains NaN values or missing data, specify the methods used to address them.

Statistical Methods (35%):

- You are encouraged to apply any of the statistical techniques covered in this course to your project.
- Appropriate selection and application of statistical techniques.
- Explanation of why certain methods were chosen.

Results and Interpretation (25%):

- Clear presentation of the results using tables, charts, or graphs.
- Accurate **interpretation** of the results and their application to society.
- Discussion of the implications of your findings.

Conclusion and Recommendations (10%):

- Summarizes key findings and insights from the analysis.
- Provides thoughtful recommendations or next steps based on the results.

Presentation (10%): The presentation will assess your ability to clearly communicate your findings and engage with the audience. You all are required to present your project to the class. The presentation will be held during week 15.

Your presentation should summarize the key findings of your project, including:

- A clear statement of the problem or research question.
- A brief explanation of the data set and methods used (e.g., statistical techniques).
- An overview of the results, with clear visuals (charts, graphs) where appropriate.
- Discussion of your conclusions, any limitations of your analysis, and potential future directions.

Format:

- You may use slides (PowerPoint, Google Slides, etc.) to structure your presentation.
- Visual aids (graphs, charts, tables) are highly encouraged.
- Make sure your slides are not overloaded with text; focus on concise points and explanations.

Time Limit:

- Each presentation should be **15 minutes or less**. There will be a 2-3 minute Q&A session following each presentation.

Presentations will be graded based on:

- Clarity and organization of the presentation.
- Use of appropriate statistical methods and interpretation of results.
- Quality and readability of visual aids (charts, graphs, etc.).
- Ability to engage the audience and answer questions effectively.

Important Project Guidelines

No complaints will be accepted within two weeks prior to the project submission date. If you are struggling to make progress or are unsure how to proceed with your project, please consult with me **well in advance**. Requests for extensions or last-minute meetings within **two weeks** of the deadline **will not be accepted**. I strongly encourage you to **start your project as soon as possible** and stop by my office for any help or guidance early in the process. The purpose of this notice is to ensure that you stay on track, understand your project requirements, and avoid starting your work only a week or two before the due date.